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(1) Research Program/Project/Study Title:	
Development of pH Indicator from Ivy Gourd (<i>Coccinia grandis</i>) Fruit	
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Program or Project has component (if no proceed to 2)? ☐ Yes ☒ No

Component Title:	
Component Leader:	
Academic Rank:	

(2) Category (please check) ☐ Social Science/Education /Humanities & other related fields ☐ Natural/Biological Sciences ☐ Food Research/ ☒ Developmental/ Computer/Engineering/ Architecture ☐ Extension ☐ Poster	(3) Research Center [Check the concerned center] ☒ SSITRDC Social Science and Industrial Technology Research Development Center ☐ LRDC Livestock Research Development Center ☐ FFRDC Fisheries and Food Research Development Center ☐ CSRDC Crop Science Research Development Center
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(4) Site of implementation	
Municipality/City:	Roxas City
Province:	Capiz
Region:	VI

(5) Classification <table> <tr> <td>Research:</td> <td>Development:</td> </tr> <tr> <td>☐ Basic</td> <td>☒ Pilot Testing</td> </tr> <tr> <td>☐ Applied</td> <td>☐ Tech. Promotion/Commercialization</td> </tr> </table>	Research:	Development:	☐ Basic	☒ Pilot Testing	☐ Applied	☐ Tech. Promotion/Commercialization	(6) Mode of Implementation ☒ Single Authorship/Proponent ☐ Collaborative Research (within campus) ☐ Collaborative Research (intercampus) ☐ Collaborative Research (external agency)
Research:	Development:						
☐ Basic	☒ Pilot Testing						
☐ Applied	☐ Tech. Promotion/Commercialization						
(7) Sector/Commodity Product Development- pH tool	(8) Discipline/Field of Specialization Chemistry						

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(9A) Abstract (This has a maximum of 250 words. It contains 1-2 sentences of rationale, followed by objectives, brief methodology, major findings, and implications of impact of the study.)

A substance property as an acid or a base is determined in terms of quantity of pH. It make use of various tools ranging from simple litmus paper to the sophisticated benchtop pH meter. Natural indicators on the other hand is utilized as an alternative to synthetic acid-base indicator. This study aimed to prepare indicator from Ivy gourd fruit and to establish its effectiveness. The experimental method of this study was composed of the preparation of raw materials, standardization of solutions, calibration, testing for effectiveness and development of color chart. The pH indicator from Ivy gourd fruit consists of Ivy gourd fruit, water, and ethanol. The process of producing pH indicator comprises of the following steps: collecting; washing; weighing; crushing; mixing with water; boiling the mixture; filtering; cooling down; mixing with ethanol; soaking; and drying. The indicator was calibrated using buffer solutions pH 0 to 14 to come up with a colored chart. This served as reference in the effectiveness tests conducted of the product. Treatment 3 of the indicator produced a distinctive colors that is comparable to the existing pH measurement tool. These finding indicates that the pH indicator from Ivy gourd fruit consists of 1:1:1 ratio of fruit, water, and ethanol can be a cheaper alternative tool in measuring pH that can be used in laboratory testing, as an instructional material and in household.

(9B) Keywords (Write 3-6 words you used in the study, preferably not found in the title because words in the title are already given keywords. This would also include process, or methodology employed.)

acid, base, natural indicator, color chart, calibration, solutions

(10) Introduction (This includes the rationale, background of the study, and objectives which are not divided by subheadings. Incorporate at least three (3) reviews)

In chemistry, a substance property as an acid or a base is only one of the main concepts that one should know. The acidity and basicity of solutions is determined in terms of quantity of pH. The “power of hydrogen” is a simple scale ranging from 0 to 14, with lower numbers than pH 7 representing higher acidity while numbers higher than pH 7 indicates basic solutions (Hren and Mikulecky, 2017). Chemistry titrations, water quality testing, testing of proper composition of biotechnology, food products preservation and tasting are few of the various activities that needs the measurement of acidity and alkalinity of a substance. It make use of various tools ranging from simple litmus paper to the sophisticated benchtop pH meter. The litmus paper is effective at indicating the acidity or basicity of a substance by changing it but cannot report an exact pH value. It works the same with hydron pH paper but the packages comes with a color-coded scale as reference when paper turns a certain color. A typical pH meter can be made of a moving coil or a digital meter, and either one or two probes. It works by the dipping the two electrodes into a solution and by ion-exchange it creates a different degree of hydrogen ion activity and the meter shows a pH measurement (Woodford, 2018). The sophisticated pH meter from the group of benchtop pH meter, wireless pH transmitters and portable pH meter offers great accuracy and convenience but costly.

Natural indicators on the other hand is utilized an alternative to synthetic acid-base indicator. This indicator are dyes and pigments that are isolated from groups of plants, fungi and algae. Plant pigment such as flavonoids with common types including anthocyanins, aurones, chalcones, flavonoids and proanthocyanidins are commonly produce in many colors in flowers and in plants such as red cabbage, berries, eggplant, turmeric, curry powder, cherries, tomato, and citrus fruits. Different studies reveal that various plants such as antirrhinum, cotton tree, marigold, dahlia, sunflower, China rose, beach morning glory, santan, fern tree, white mulberry and promeganate show good acid base indicator activity against

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various synthetic pH indicators (Garg, et al, 2013). On the other hand, plant parts such as flowers and leaves containing pigments such as flavonoids and anthocyanins are utilized as natural indicators (Fuentes, et al, 2021). Various plants show good acid-base indicator activity such as white mulberry and pomegranate (Garg, et al, 2018), red cabbage, berries, eggplant, turmeric, okra and fluted gourd (Abuh, 2018), gumamela (Gupta, Jain & Jain, 2012), dragon fruit (Azlim, et al, 2022), and grape and spinach (Veiga-Santos, Ditchfield, and Tadini, 2011).

In the Philippines, local plants are used for acid-base testing: 2-2021-050862 *Phyllanthus Phyllanthus multiflorus Willd* offers distinct color change in different solutions, thus, comparable to commercial pH paper (Fuentes, Legaspi, and Albaladejo, 2021); 2-2015-000566 *Kamias Averrhoa bilimbi L.* Flower Extract can be used to determine the pH of a sample formulation using an established color chart (Jusayan, 2015); 2-2017-000398 Butterfly pea *Clitoria tematea Linn.* flower commonly known as blue turnate extract serves as indicator of the acidity and alkalinity level of the solution (Facultad & Descarial, 2018); 2-2017-000623 Ethanol extract of pandan *Pandanus Amaryllifolius* leaves and gumamela *Hibiscus Rosasinensis* flower can also be used as natural indicator (Mangubat & Mangubat, 2018). An acid-base indicator was derived from Moses-in-a-Basket *Tradescantia Spathacea* petroleum ether leaf extract (Echapore & Mendaño, 2018).

The Philippines has plenty of plants and weeds that can be developed as natural indicator. The Ivy Gourd, also known as Baby Watermelon is a fast growing tropical vine and native in the Philippines. It grows aggressively and smother and kill native vegetation as it climbs and covers shrubs and trees thus prevents growth. It has ovoid to elliptical fruit that turns red when ripe. The fruits of *Coccinia grandis* are recognized to have active constituents similar to amyran, lupeol, taraxerone, taxerol, glycoside cucurbitacin B, and secondary metabolites like alkaloids, flavonoids, saponins, glycosides (Niranjan, 2010). The fruits are traditionally used for culinary and medicinal purposes. It is considered as one of the most beneficial medicinal herbs in traditional and ayurvedic medicine as it contains pharmacological actions such as antibacterial and antimicrobial (Bhattacharya et al., 2010), antioxidant (Deshpande et al., 2011), antiulcer (Girish, C & Vineela, S & NarasimhaReddy, Y & Reddy, O.V.S. & Rajasekhar, K.K. & Shankarananth, 2011), hypoglycemic & antidiabetic (Islam et al., 2011), anticancer (Bhattacharya, 2011), anthelmintic (Tamilselvan N, Thirumalai T, Elumalai EK, Balaji R, 2011; Dewanjee et al., 2007); analgesic (Pattanayak, 2009), hepatoprotective (Vadivu, 2008), and anti-inflammatory (Deepti et al., 2012). However, in this study, the potential of Ivy gourd fruit as pH indicator was explored.

Thus, the researchers aimed to prepare acid-base indicator from Ivy gourd fruit extract. Specifically, this study aimed to:

1. prepare indicator from Ivy gourd fruit ethanol extract; and
2. establish the effectiveness of the developed indicator.

(11) Materials and Methods or Methodology (This includes the materials, methods, duration & location of the study. Materials and Methods are headings used in Natural Science researches and methodology is Social Sciences. Presentation of materials may not necessarily be in separate heading but may be incorporated in the specific methods used. Justify paragraphs with main headings in caps.)

Materials

The materials used in this study are the following: Ivy gourd fruit, water, ethanol, filter paper, different acid and base solutions, hydron paper, portable pH meter, laboratory thermometer, glass rod, watch glass, glass funnel, graduated cylinder, and beaker.

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Methods

This study utilized the experimental method composed of the preparation of raw materials, standardization of solutions, calibration of Ivy gourd fruit ethanolic extract, testing for effectiveness and development of color chart.

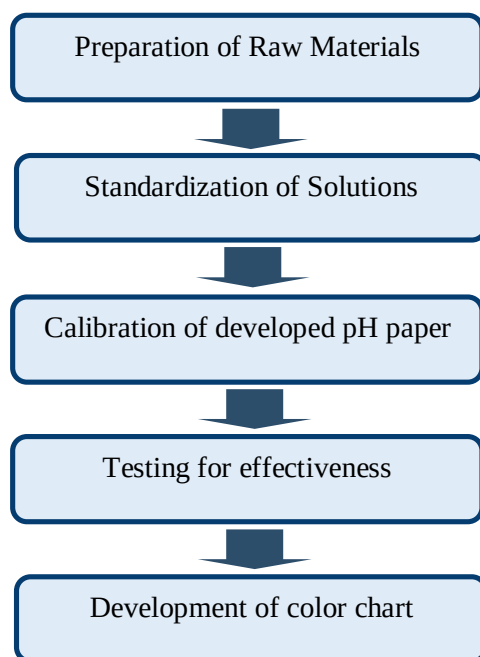


Figure 1: Process flow

Preparation of Raw Materials. This includes the steps of:

- collecting of ripe ivy gourd fruit (bright red)
- washing of collected ripe ivy gourd fruit using tap water
- cutting of ivy gourd fruit lengthwise;
- weighing of 30g of Ivy gourd fruit
- measuring of water - 30 ml of distilled water were used;
- mixing- cut ivy gourd fruit were mixed with 30mL water;
- boiling- the mixture were boiled to 150 °F to 180 °F until 25% of its original volume remains to produce a darker color of extracts;
- mashing- the boiled Ivy gourd fruit were mashed to produce extracts;
- extracting fruit extract;
- filtering- to separate solid particles, the boiled mixture was filtered using a muslin or cheesecloth. The remaining residues were further squeeze to produce more extracts;
- cooling- the filtrate were cooled down to 104 -112 °F;
- mixing the filtrate to 30 mL of 75% ethanol;
- soaking the filter paper strips- prepared filter paper strips cut into 50mm x 6mm were soaked to the mixture for 120 seconds. This is to give more time for the extracts to penetrate to the filter paper since extracts produce were pale in color.
- drying- soaked filter paper were blow dried at 284 °F.

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Composition Formulation. Composition was determined by comparing three different formulations in three replications as shown in the table below.

Table 1: Treatments of the composition of natural pH indicator

Composition	1 st Formulation	2 nd Formulation	3 rd Formulation
Ivy gourd fruit	30g	30g	30g
Water	20 mL	25mL	30mL
75% ethanol	30mL	30mL	30mL

Standardization of Solutions. Preparing different solutions pH 0 to 14 using the portable pH meter.

Table 2: pH levels of different solutions

Solutions	pH
Hydrochloric acid	1
Acetic acid	2
Carbonic acid	3
Tomato juice	4
Black Coffee	5
Milk	6
Distilled water	7
Sea water	8
Sodium bicarbonate	9
Milk of magnesia	10
Dishwashing Liquid	11
Ammonia solution	12
Sodium hypochlorite	13
Sodium Hydroxide	14

Calibration of developed indicator. Calibrating and testing the effectiveness of the indicator comprises of the following steps:dipping of individual strip of paper in each buffer solutions, and comparison to litmus paper, Hydron paper, portable pH meter results.

Development of color chart. The color chart is based on the best formulation of the study that produced distinctive colors for pH 1-14.

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(12) Results and Discussions (This part contains presentation of results (data) with reference to the table and figure, to be followed by discussion, with support of reviewed literature. Sequence shall be based on the objectives of the study. Tables are formatted without the vertical lines. Horizontal lines are set at the top and bottom portions as well as below table headings and above major subdivisions. Justify paragraphs with main headings in caps.)

Development of Color Chart

The best formulation of the study that produced distinctive colors for pH 1-14 comprises 30g ivy gourd fruit, 30 mL water and 30 mL 75% ethanol as shown in the table below.

Table 3: Color reaction of different formulations.

Color Change			
pH Level	1 st Formulation	2 nd Formulation	3 rd Formulation
1			Red
2			Dark Violet
3			Violet
4			Light Violet
5			Pink
6			Light Pink
7			White
8			Dark Blue
9			Dark Green
10			Blue Green
11			Yellow Green
12			Light Green
13			Green
14			Yellow

The color chart developed of the natural indicator from Ivy gourd fruit is shown below.

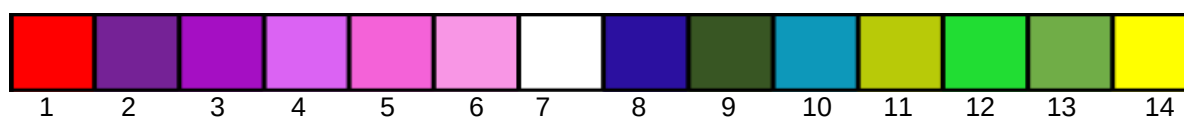


Figure 2: Developed color chart.

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Comparison to other measurement tools

Color reaction of treatment 3 were compared with the different pH measurement tools including the litmus paper test, Hydron paper, and portable pH meter. Results showed that the prepared indicator is comparable with other types of pH measurement tool in terms of color reaction.

Table 4. Comparison of results of different pH measurement tools

Solutions	Color Reaction				Portable pH meter
	Litmus paper		Hydron paper	Ivy gourd Indicator	
	Red	Blue			
Hydrochloric acid	Red	Red	Dark Red	Red	1
Acetic acid	Red	Red	Red	Dark Violet	2
Carbonic acid	Red	Red	Orange	Violet	3
Tomato juice	Red	Red	Light Orange	Light Violet	4
Black Coffee	Red	Red	Yellow Orange	Pink	5
Milk	Red	Red	Light Brown	Light Pink	6
Distilled water	Red	Blue	Dark Yellow	White	7
Sea water	Blue	Blue	Yellow Green	Dark Blue	8
Sodium bicarbonate	Blue	Blue	Green	Dark Green	9
Milk of magnesia	Blue	Blue	Dark Green	Blue Green	10
Dishwashing Liquid	Blue	Blue	Dark Blue	Yellow Green	11
Ammonia solution	Blue	Blue	Blue Violet	Light Green	12
Sodium hypochlorite	Blue	Blue	Blue Green	Green	13
Sodium Hydroxide	Blue	Blue	Yellow	Yellow	14

(13) Conclusions and Recommendations (This part shall be written in narrative form and arranged based on the objectives and findings. Conclusions are general statements answering the objectives. Recommendations are based on the conclusions. They may be split or combined based on their length, which means, conclusions and recommendations may be split under separate headings if they are long. Justify paragraphs with main headings in caps.)

Based on the results of the study, the following conclusions were drawn:

1. Ivy gourd fruit can be used as a natural indicator; Treatment 3 of the prepared indicator comprises of 30g ivy gourd fruit, 30 mL water, and 30 mL 75% ethanol can be used as a natural pH indicator.

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2. The natural indicator from ivy gourd fruit is comparable with the existing pH measurement tool; and the natural indicator from ivy gourd fruit can be an effective alternative tool for measuring pH.

However, it is recommended that acceptability testing of experimentatlists and science educators should be conducted, and shelf-life of the indicator should be tested. Also, the experimental component of this research be continued for the refinement of the color chart as reference for future analysis.

(14) Output (Indicate the specific products, processes or services which the project has produced; quantify when possible.)

Output: Natural pH indicator from Ivy Gourd (*Coccolinia grandis*) Fruit and the process of producing thereof

(15) Technology Transfer Activity (Indicate the specific products, activity design, processes or services transferred as extension activities)

N/A

(16) Economic Analysis (This should present the direct cost of project implementation, expenses in the production of technology, revenues generated from technology produced, return on investment (ROI) and payback period.)

Direct Cost

Travel Expenses (Local)	P 200. 00
Supplies and Materials/Raw Materials	P 1, 500.00
Tools	P 2, 500.00
Facility and Equipment Rental Expenses	P - -
Documentation Printing, Photocopying Expenses	P 300.00
Total	P 4,500.00

ROI (%) = Net Income /Total Project Cost

$$= 2,100/4500$$

$$= 47\%$$

Payback Period (Years) = Total Project Cost /Average Annual Profit

$$= 4500/5,160$$

$$= 0.9$$

(17) Acknowledgments (Include only important persons and institutions with involvement in funding and conduct of the research. Justify paragraphs with main headings in caps.)

Sincere gratitude to the following who helped in the completion of this study:Dayao Satellite College Administration., for the permission of the conduct of the study at the Physico-Chemical laboratory room and CapSU Dayao Research Office.

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(18) References (It should include only works mentioned in the text and should be arranged alphabetically. Citations shall not be less than 13 entries. This must follow the APA latest edition format, in hanging indentation with 12-point space in between.)

Abuh, L. (2018). Ethanolic extract of *Telfairia Occidentalis* and *Abelmoschus osculentus* flowers as substitutes to hazardous synthetic indicators in acid-base titration. *International Journal of Chemical Science Online* ISSN: 2523-2843 Volume 2. Page 16-19

Azlim, N. A., Mohammadi Nafchi, A., Oladzadabbasabadi, N., Ariffin, F., Ghalambor, P., Jafarzadeh, S., & Al-Hassan, A. A. (2022). Fabrication and characterization of a pH-sensitive intelligent film incorporating dragon fruit skin extract. *Food Science & Nutrition*, 10, 597– 608. <https://doi.org/10.1002/fsn3.2680>

Deepti, B., Sasidhar, K., Sadhana, G. and Thaakur, S., (2012). Wound healing activity of chloroform extract of *Coccinia grandis* on excision, incision & dead space wound model in rats. *International Journal of Research in Pharmaceutical Sciences*, 3, pp.470-475.

Deshpande SV, Patil MJ, Daswadhkar SC, Suralkar U, Agarwal A (2011). A study on anti-inflammatory activity of the leaf and stem extracts of *Coccinia grandis* Voigt. *Int J Appl Bio Pharm*. 2011b;2:247–250.

Dewanjee, S., Maiti, A., Kundu, M. and Mandal, S.C., 2007. Evaluation of anthelmintic activity of crude extracts of *Diospyros peregrina*, *Coccinia grandis* and *Schima wallichii*. *Dhaka University Journal of Pharmaceutical Sciences*, 6(2), pp.121-123.

Echapare, E. & Mendaño, R. (2018). ACID-BASE INDICATOR FROM MOSES IN A BASKET (TRADESCANTIA SPATHACEA) LEAF EXTRACT (PH 2/2018/001206). Intellectual Property Office of the Philippines. https://patentscope.wipo.int/search/en/detail.jsf?docId=PH234859061&_cid=P11-L2K7WM-39770-1

Facultad, MJ & Descarial, M (2018) A PROCESS OF PRODUCING NATURAL ACID-BASE INDICATOR (PH 2/2017/000398).). Intellectual Property Office of the Philippines. https://patentscope.wipo.int/search/en/detail.jsf?docId=PH278362979&_cid=P11-L2K7O6-35853-1

Fuentes, P., Legaspi, J. and Albaladejo, V. (2021). *Phyllanthus* (*Phyllanthus multiflorus* Willd.) Fruit as Natural pH Indicator. *International Journal of Advanced Engineering Research and Science*, 8(10), pp.335-339.

Garg, S., Garg, A., Shukla P., Mathew, J., Vishkarma, D. & Sen, S. (2018). Investigation of Plant Source as Natural Indicators for Acid Base Titration to Reduce the Use of Harmful Chemicals in Some Extent. *International Journal of Biology, Pharmacy and Allied Science* ISSN: 2277-4998. January, 2018, 7(1): 64-70

Girish C, Balakrishnan S, Koner BC, Jayanthi S, Rao KR, Rajesh B, Pradhan SC. (2011). Evaluation of antiulcer activity of *Coccinia grandis* leaves. *Res J Pharmacology & Pharmacodynamics*. 2011;3:2011–2013.

Gupta, P., Jain, P., & Jain, P.K. (2012). Isolation of natural acid base indicator from the flower sap of *Hibiscus rosa sinensis*. *Journal of Chemical and Pharmaceutical Research*, 4(12):4957-4960

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Islam MS, Parvin S, Uddin MN, Mazid MA. (2014). Antidiabetic and antioxidant activities of decoctions of *Coccinia grandis* Linn. and *Centella asiatica* (L.) on alloxan-induced diabetic rats. *Bangladesh Pharm J*; 17(1):86–91.

Jusayan, Shirley R. (2015). pH Indicator from Kamias (*Averrhoa bilimbi* L.) Flower Extract. *WVSU Research Journal* Vol. 4 No. 1 PP. 30-40

Mane, A.N., Koli, D.S., Kumbhar, V.B. (2016). Use of *Combretum Indicum* Flower Extract as a Natural Indicator in Acid-Base Titration. *International Journal of Institutional Pharmacy and Life Sciences* 6(3): May-June 2016. ISSN: 2249-6807

Mangubat, F. & Mangubat, F (2018) ACID-BASE INDICATOR FROM PANDAN (*Pandanus amaryllifolius*) LEAVES AND GUMAMELA (*Hibiscus rosasensinsis*) FLOWER (PH 2/2017/000623). Intellectual Property Office of the Philippines. https://patentscope.wipo.int/search/en/detail.jsf?docId=PH278364989&_cid=P11-L2K7S8-37467-1

Pattanayak PS, Priyashree S. (2009). In vivo antitussive activity of *Coccinia grandis* against irritant aerosol and sulfur dioxide induced cough model in rodents). *Bangladesh J Pharmacol*; 4: 84-7

Sutar Niranjan , Ranju Garai, Umesh Kumar Sharma, Uma Shankar Sharma, Gautam Kumar Jana, Abhishek Singh. (2010). Evaluation of Anti-inflammatory Activity of *Coccinia indica* Leaves Extracts, *Journal of Pharmacy Research*, Vol 3(9), Pp 2172-2173.

Tamilselvan N, Thirumalai T, Elumalai EK, Balaji R, David E. (2011). Pharmacognosy of *Coccinia grandis* - a review. *Asian Pac J Trop Biomed*. 2011;1:S299–S302.

Vadivu R, Krithika A, Biplab C, Dedegpya P, Shoeb N, Lakshmi KS. (2008). Evaluation of hepatoprotective activity of the fruits of *Coccinia grandis* (Linn.) *Int J Health Res*. 2008;1:163–168.

Veiga-Santos, P., Ditchfield, C. and Tadini, C.C. (2011), Development and evaluation of a novel pH indicator biodegradable film based on cassava starch. *J. Appl. Polym. Sci.*, 120: 1069-1079. <https://doi.org/10.1002/app.33255>

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